

Intro to Technical Analysis

A Guide to Understanding Technical Analysis and its Applications

RJOFutures

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Introduction

Thank you for your interest in RJO Futures *Intro to Technical Analysis*.

Trading futures has the potential to be very rewarding. However, it can take hard work and a lot of time to keep up with changing markets and trends. As a trader, one way to even the playing field somewhat is the use of technical analysis as part of your trading plan. Technical analysis can be a useful, customizable tool to help give you an edge. This guide is meant to give novice traders a solid start to understanding technical analysis and how to apply it. Additionally, it can be used by more experienced traders as a refresher course. This guide was written by RJO Futures' trading strategists, applying their many years of industry knowledge and experience. As you study the content, we encourage you to contact us at any time with questions or comments. It is our goal to help you understand and apply the information herein.

Regards,

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The risk of loss in trading commodity futures and options is substantial. Before trading, you should carefully consider your financial position to determine if futures trading is appropriate. When trading futures and/or options, it is possible to lose more than the full value of your account. All funds committed should be risk capital. Past performance is not necessarily indicative of future results.

Table of Contents

PART I: THE W'S OF TECHNICAL ANALYSIS.....	3
WHO Uses Technical Analysis?.....	3
WHY Is Technical Analysis Important?	3
WHAT Are Charts?	4
WHEN Do You Use Charts?	4
WHERE Can You Find Charts?	4
PART II: CHART PATTERNS.....	6
Trendlines.....	6
Channels	9
Continuation Patterns.....	10
<i>Bull/Bear Flags & Pennants</i>	10
<i>Triangles</i>	11
<i>Rectangles</i>	13
<i>Wedge</i>	14
Reversal Patterns.....	15
<i>Head & Shoulders</i>	15
<i>Tops & Bottoms</i>	16
<i>Island Reversals</i>	18
Fibonacci Retracement Numbers.....	19
PART III: CANDLESTICKS	20
Doji	20
Gravestone Doji	21
Long-Legged Doji	22
Bullish & Bearish Engulfing Patterns	23
Dark Cloud Cover	23
Morning Star	24
Shooting Star	25
PART IV: USING AND APPLYING STUDIES.....	26
Moving Averages	26
Bollinger Bands.....	26
Relative Strength Index (RSI).....	26
Moving Average Convergence/Divergence (MACD)	27
Momentum.....	27
Williams %R.....	27
Volume	27
Open Interest	27
Commitment of Traders (COT).....	27
Average Decline Index (ADI)	28
Stochastic Fast & Slow	28
PART V: MORE INFORMATION ABOUT TECHNICAL ANALYSIS.....	32

Part 1: The W's of Technical Analysis

WHO USES TECHNICAL ANALYSIS?

If you are a trader you will likely fall into one of three categories—*technicians*, *fundamentalists*, or a combination of the two. Technicians rely solely on historical chart patterns to predict future price movements. Fundamentalists rely on real-life events that may drive a market such as the increasing demand for corn to produce ethanol. Many people rely on both technical and fundamental information to formulate market opinions as well. To trade successfully, we try to make an educated decision to buy and sell in a timely fashion. Even if the fundamental information is bullish or bearish, how do you know that the timing to make the trade is right to initiate or liquidate a position? Studying and executing good technical analysis practices can help you improve your odds of success.

Anyone that trades the futures market must also learn to use a trading plan that complements their lifestyle. Trading can be very stressful, especially if you are in a losing trade. Completely changing your lifestyle to trade can add to the stressfulness of trading. Therefore, once you figure out if you are a technician, fundamentalist, or a combination of the two, you should decide if you are a *short-term*, *intermediate-term* (*swing trader*), or *long-term position trader*. Short-term traders typically hold positions for 1 to 3 days; intermediate-term traders typically hold positions for 3 days to 3 weeks. Long-term traders hold positions for more than 3 weeks. You never want to over leverage yourself when trading futures. The longer you hold positions, the more money you should be willing to risk on a trade to withstand the day-to-day market volatility.

Market participants are typically broken down into *small speculative traders*, *large speculative traders (funds)*, and *commercial hedge traders*. Each exchange submits a Commitment of Traders report weekly with the total long and short positions for each group. This information can be useful in conjunction with technical analysis to improve your timing of entering and exiting the market. For example, if the funds and small specs both hold a record net long position, at some point the market will likely experience profit taking and reverse. However, it is important to keep in mind the old saying that “records are meant to be broken.”

WHY IS TECHNICAL ANALYSIS IMPORTANT?

According to John J. Murphy's book, *Technical Analysis of the Futures Markets*, there are three basic reasons why technical analysis is such a popular tool for analyzing the markets. According to the book, these include: market action, trends in price moves, and the fact that history repeats itself.

With the speed of information changing hands globally these days, there are many factors that can quickly change a market—such as a surprise interest rate change or political turmoil in a major energy producing country. By being aware of key areas in the market, you can work to better manage your risk and take advantage of market breakouts. Always remember the old adage: “the trend is your friend.” Following trends is not a new approach. If you trade with the trend, your odds of success are likely to increase as a result. Understanding technical analysis can help you with identifying the trend as well as timing the trade. When you are successful with timing the trade, you will likely become more psychologically disciplined as a trader.

WHAT ARE CHARTS?

Charts are a graphical display of historical market data. It is common to look at charts on both shorter-term and longer-term scales depending on your style of trading. Charts are available as intraday, daily, weekly, and monthly. Typically, shorter-term traders use shorter-term time intervals on charts to analyze market data. Longer-term traders tend to use longer-term time intervals on charts to analyze data. The most common types of charts—Bar Charts, Line Charts, and Candlestick Charts—are all available in the RJO Vantage platform. For the purpose of differentiating the types of charts, we have attached the Daily Chart for the e-CBOT Corn contract as a bar chart, line chart, and candlestick chart on page 6.

WHEN DO YOU USE CHARTS?

Anyone that has traded can confirm that timing on entering and exiting a market is critical to your long-term trading success. By using technical analysis, you can work to place buy stops above the resistance lines and place sell stops below support lines to either enter or exit a market. By having a deeper understanding of technical analysis, you will likely be able to make better decisions to hold or liquidate trades as well. If you would like more information on money management, call your RJO Futures representative to request the *Risk Management Guide*.

WHERE CAN YOU FIND CHARTS?

Web

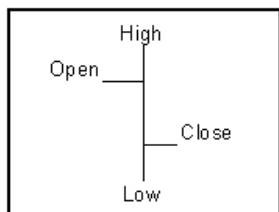
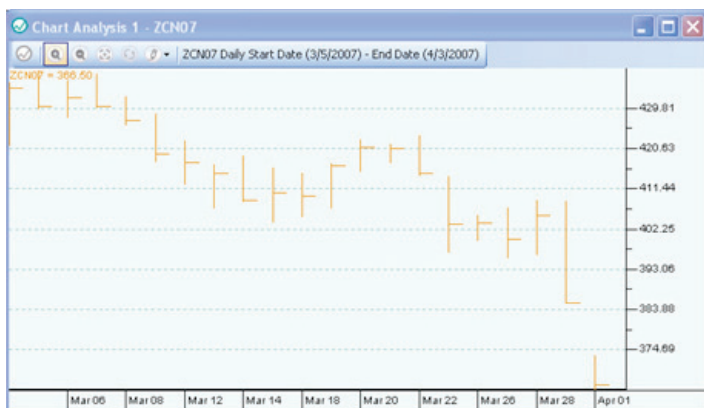
RJO Futures offers free delayed charts at www.rjofutures.com. Our free charts area includes advanced study filters, so you can view charts in your preferred style. An easy “Symbol Lookup” section with “Contract Details” button is also provided. If you prefer real-time, online chart data, you can subscribe to various fee-based chart packages through websites such as www.barchart.com or www.futuresource.com.

Trading Software

RJO Futures offers clients real-time charts through our proprietary electronic trading platform, RJO Vantage™. Or you can access charts through our Web-based trading platform, Web OE. Both platforms are free to use for RJO Futures clients. Download the latest version of RJO Vantage™ or receive access to Web OE by clicking “Trade Futures/Forex”, then “Online Trading Platforms” in the top navigation bar at www.rjofutures.com.

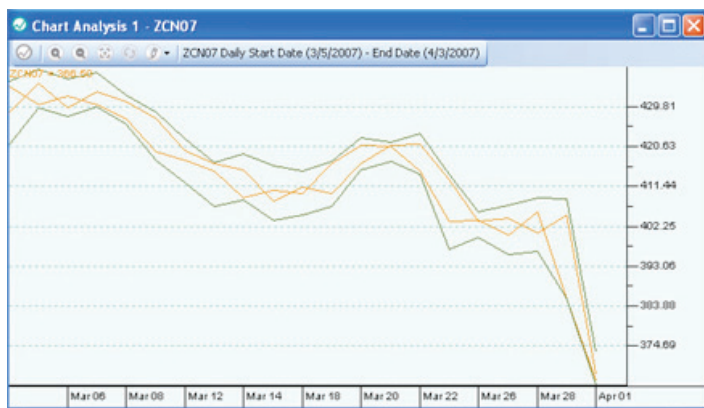
Chart Types

BAR CHARTS



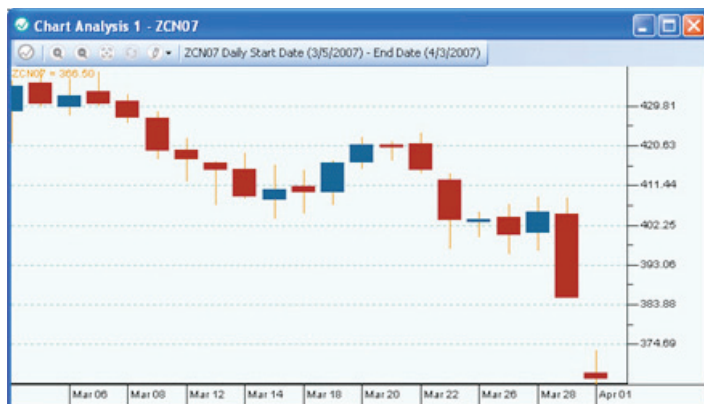
Each bar represents the price action based on the viewed time scale.

LINE CHARTS



Each line represents the Open, High, Low, or Close of the time intervals.

CANDLESTICK CHARTS



Each bar can represent a minute, day, week, or even month, but the chosen time frame does not influence the color of the candle because a hollow bar will always be created when the close is higher than the open.

Part 2: Chart Patterns

This section aims to identify some of the most common patterns for identifying market direction and target prices.

TRENDLINES

The chart patterns are important to help you recognize market changes. First and foremost, trendlines are a valuable tool used in recognizing bullish, bearish, or sideways trends. Trendlines are usually studied in three time frames including short term, intermediate term, and long term. The longer the market has been in a trend, the more significant the trendlines are. To recognize bull and bear markets, look for bullish trends to consist of higher highs and higher lows and bearish trends to consist of lower lows and lower highs. If a bullish trend sells off below the most recent low (and vice versa on a bearish trend), the most simplistic definition of the trend is violated. This is also known as market divergence and is often viewed as the first clue that a trend may be changing.

Trendlines moving upward should be viewed as **support** (or a floor) until broken. Likewise, trendlines moving downward should be viewed as **resistance** (or a ceiling) until broken. From a risk management perspective, sell stops should be placed below support and buy stops should be placed above resistance. How far above resistance or below support should you place your buy or sell stops? This might depend on the market volume and volatility that you are working to place a stop in. Discuss this further with your RJO Futures advisor.

Another useful tool to look at for support and resistance is the volume traded. Higher volume traded when the support or resistance levels are held indicates that the trendline is more significant. In pit traded markets, volume is posted with a one-day lag. Electronic market volume is available the same day and is updated as contracts trade.

Another common tool used to help identify the trend is the simple moving average. The moving average gives you a smoother view of the trend and any market changes. You can customize the moving average to fit your needs by setting up the computer to allow for a specific number of days to calculate it. The most common moving averages used include the 4-, 9-, and 18-day periods. For example, a 9-day moving average would calculate the closing prices from the most recent 9 periods of trading on the chart. The moving average study can be used on any chart from short-term intraday charts to longer-term monthly charts. If you use the 9-period moving average on a 60-minute bar chart, the moving average would be calculated from the most recent nine 60-minute bars.

The exponential moving average is also a common tool. It is similar to the simple moving average, but is more sensitive to shorter time frame market changes than the simple moving average. Therefore, you might prefer the exponential moving average over the simple moving average.

Identifying true trend changes early can help you improve your timing of entering or exiting the market. Many factors come into play when identifying a trend change. Most notably:

- The market closes above the resistance or below the support trendline.
- Volume tends to increase as a trendline is broken.
- The simple moving average changes direction or the exponential moving average lines crossover.

There are many theories to try to help traders determine if the violation of the trendline is genuine. According to *Technical Analysis of the Futures Markets* by John J. Murphy, the most common theories are the 3% price move through the long-term trends, 1% price move on shorter-term trends, and/or the 2-day rule. The 2-day rule implies that the market should hold the breakthrough for at least 2 trading days to confirm a trend change.

Bullish Trend Chart



Bearish Trend Chart

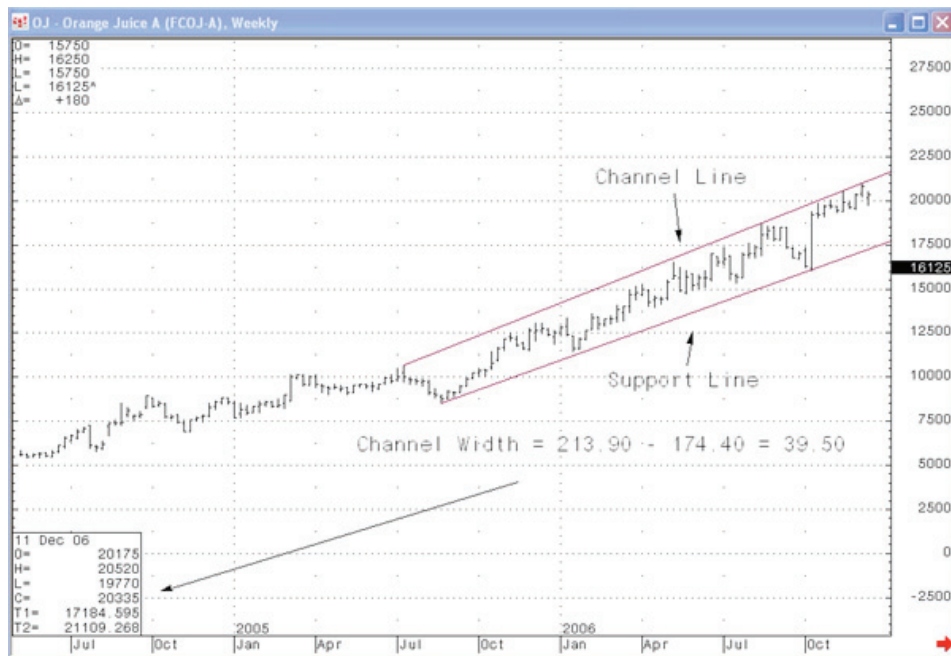


CHANNELS

Channels are very similar to trendlines. The basic characteristics of a trendline are carried through when we look at channels. Rather than just drawing support or resistance, we also draw an uptrending line or downtrending line to signify a range that the market will likely trade in. Channels are commonly used by more aggressive traders looking to try to capture near-term or countertrend market swings. If you are not an aggressive trader, channels are commonly used to try to give warning to an approaching trend change. As a trend has run its course, the market struggles to reach the channel lines. In the chart below, the chart fails to rally to the uptrending channel line from the end of 2006 through early 2007.

According to John J. Murphy, *"Once a breakout occurs from an existing price channel, prices usually travel a distance equal to the width of the channel."* In this case, the width of the channel is equal to 39.50. The market broke to the downside at 195.60. Therefore, we would expect the market to have a near-term downside target of 156.10.

Weekly Chart - FCOJ



CONTINUATION PATTERNS

Probabilities are higher that a market trend will continue in the same direction upon completing one of these patterns. Keep in mind that these patterns are not foolproof. They can catch you off guard with a trend reversal.

BULL/BEAR FLAGS & PENNANTS

Flags and pennants are both very similar in the futures markets. They are created as a result of a sharp high-volume market move and the need for the market to take a breather. Both flags and pennants usually last one to three weeks before the longer-term trend resumes. The flags and pennants are usually not as long lived in bear markets compared to bull markets. Once the flag and pennant formations have completed, the market typically resumes the original trend with higher volume. The flag is a parallelogram that countertrends the longer-term trend. Additionally, the pennant resembles a symmetrical triangle with a much larger rally or sells off upon breakout. The pennant move is likely to be equal to the trend move prior to the pennant formation.

Technicians often look for flags because they are viewed simply as a breather from a highly volatile market situation with high odds of continuing the trend. Flags are often referred to as a highly predictable continuation pattern. In a bull market, look for a short-term downtrending channel prior to a continuation of the uptrending market. In a bear market, look for an uptrending short-term channel prior to continuing the down trend.

Bull Flag Example



TRIANGLES

Symmetrical Triangle

Symmetrical triangles typically represent a pause in the bigger trend with lower and lower volume as the triangle is formed. Both the lower and upper boundaries of the symmetrical triangle are converging equally. The resistance line of the triangle is downward sloping while the support line of the triangle is upward sloping. Although it is not a guarantee, this is considered to be a continuation pattern because the odds are in favor of the market breaking out in the direction of the longer-term trend. If the market was in a bull market prior to the symmetrical triangle being formed, it will likely breakout to the upside. If the market was in a bear market prior to the symmetrical triangle being formed, it will likely breakout to the downside. As the triangle breakouts, the volume increases as well. The symmetrical triangle allows the trader to narrow the timing of a breakout. The longer the market trades in the triangle, the greater the odds a breakout is approaching since the breakout should occur before the apex of the triangle is reached.

Once the price breaks out of the triangle, look for a market movement equal to the distance between the widest points of the triangle.



Ascending Triangle

Ascending triangles are considered bullish patterns. The upper trendline is flat while the lower trendline is pointing up. Many of the characteristics of the symmetrical triangle hold true for the ascending triangle. Most notably, the volume increases significantly on a breakout to the upside.



Descending Triangle

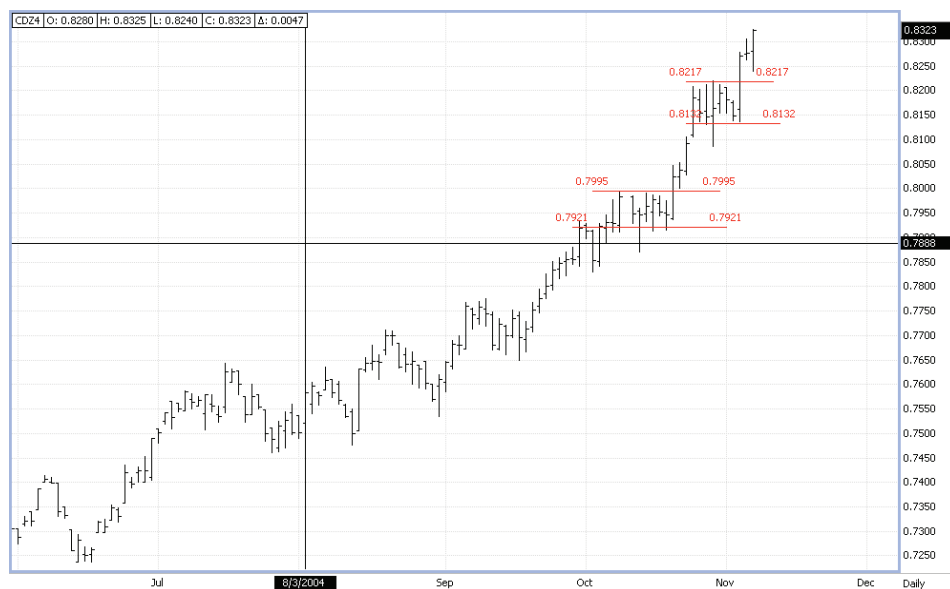
Descending triangles are considered bearish patterns. The upper trendline is descending while the lower trendline is flat. Many of the characteristics are similar to that of the symmetrical triangle. Most notably, the volume increases significantly on a breakout to the downside.



RECTANGLES

Rectangles are similar to the symmetrical triangle except for the fact that the upper and lower trends run parallel to each other. (It looks like a short-term sideways market.) They are typically considered to be continuation patterns because the odds of a breakout in the direction of the longer-term trend are high. However, it is possible that the rectangle acts as a reversal pattern. The volume remains higher in rectangle formations than during the formation of triangles due to the wide price swings. Even though the volume remains high during the formation of the rectangle, volume still increases on a breakout. The market bounces between the upper and lower portions of the rectangle usually for 1 – 3 months until the market breaks out one way or another.

Volume is the key to predicting which way the market will likely breakout when it does. If volume is higher on the up trending swings versus the down trending swings, the market will likely break to the upside and vice versa.



WEDGE

Once again, the wedge is similar to the symmetrical triangle except both sides of the triangle are either pointing up (rising) or pointing down (falling). Also, the formation tends to breakout much closer to the apex than in a symmetrical triangle. As with other triangle formations, you should notice a drop in volume as the pattern forms and increases upon breakout. A falling wedge is considered bullish. A rising wedge is considered bearish. Wedges are most commonly seen as a continuation pattern of the longer-term trend. However, they are found occasionally in trend reversal situations.



REVERSAL PATTERNS

Probabilities are higher that a market trend will change directions upon completing a reversal pattern. Keep in mind that these patterns can fail though.

HEAD & SHOULDERS

Continuation gaps occur when the price action is breaking out of their trading range or congestion area. Prior to a continuation gap a market will congest for a period of time ranging from days to weeks. The area near the top of the congestion area is usually resistance when approached from below. Likewise, the area near the bottom of the congestion area is support when approached from above. To break out of these areas requires market enthusiasm and either many more buyers than sellers for upside breakouts or more sellers than buyers for downside breakouts. Increased volume should accompany the market as it gaps. Don't fall into the trap of thinking that this type of gap, if associated with good volume, will be filled soon.

	H&S	Inverted H&S
Volatility	High	Lower typically
Duration	Shorter	Longer



TOPS AND BOTTOMS

Tops & bottoms form as the market is preparing for a major trend change. Volatility tends to increase as a result of the market playing tug of war to find the fair trade prices in correlation to supply and demand.

Double Tops & Bottoms

Double tops & bottoms are an excellent indicator of a market that is ready for a trend change. These are much more common than triple tops. In tops, volume should be lower on the second leg up and should increase when the low between the peaks is broken. The opposite is true for a double bottom. Volume should decrease on the second leg down and increase when the high between the valleys are broken. You increase your odds of success by waiting to enter the market until the trend has changed.

Double Top Chart

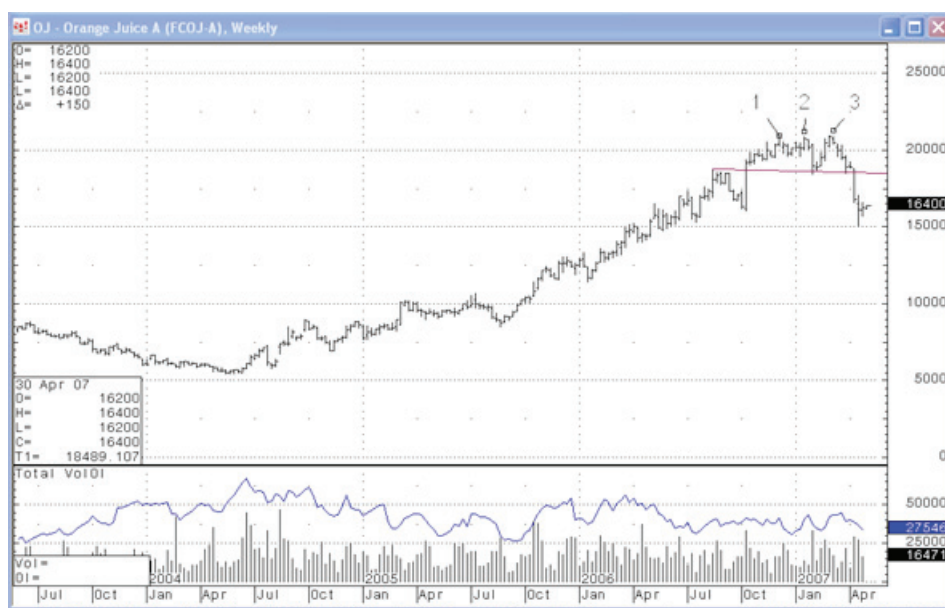


Double Bottom Chart



TRIPLE TOPS & BOTTOMS

Triple tops & bottoms are almost exactly the same as the head & shoulders (tops) and inverted head & shoulders (bottoms) patterns except they are more of a sideways channel. In the triple top or triple bottom, we are looking for three peaks or valleys that are similar in price followed by a trend change. The most important factor to watch for is the volume. Volume must increase during a breakout from the neckline. As with all trendlines, the more often a price fails to break through a congestion area, the more significant the support or resistance become.

Triple Top Chart

ISLAND REVERSALS

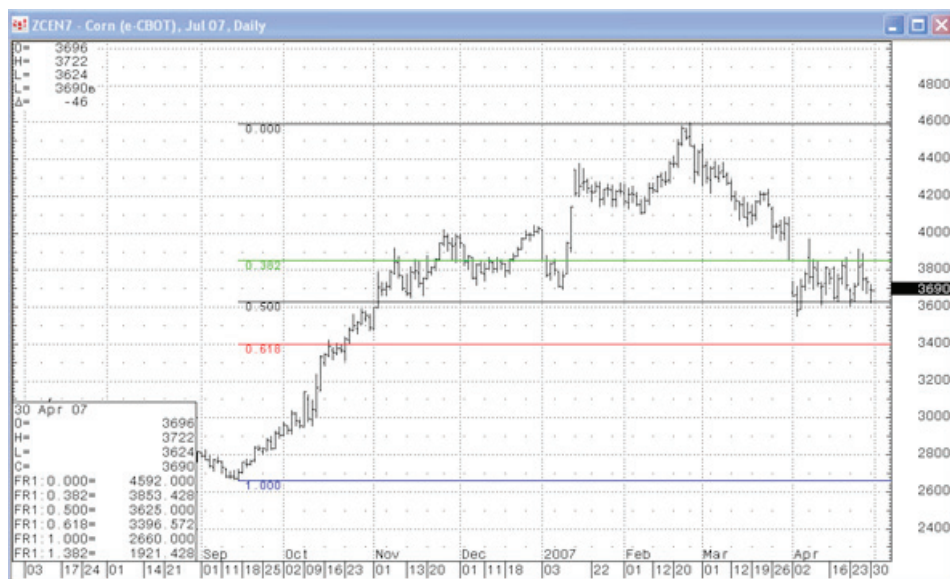
Island reversals usually occur when there is a major trend change from bullish to bearish. After an exhaustion gap occurs, the market will trade in a minor sideways channel for a few days then gap lower. It will then have a breakaway gap to the downside that will leave behind what appears to be an island. The exhaustion gap is only labeled so when it fails to hold the upside momentum and corrects leaving a top in the market.



FIBONACCI RETRACEMENT NUMBERS

When analyzing the markets it is also important to address how far a market will move in order to work profit objectives and manage risk. Whether your analysis indicates that the market trend will continue or reverse, your bottom line is affected. In many of the previous examples, we addressed the minimum objective of each move based on the distance of the channel, the widest point of the triangle, or the distance between the head and neckline in the head & shoulder formation.

Fibonacci retracement numbers are commonly used to look at target points for reversals or corrections in the market. Identifying the strength of the initial trend may help you analyze the correction. It is commonly believed that markets tend to retrace 38.2%, 50%, 61.8% or 100%. 38.2% (1-.618) is actually the inverse of the 61.8% retracement level. It is believed that stronger trends will only correct 38.2% while weaker trends have much stronger corrections of 61.8%.



Part 3: Candlesticks

Candlestick charting is a Japanese method of tracking the rise and fall of prices. Called Candle Chart Analysis since the lines mimic the shape of a candle, these charts provide a simple way of looking at prices and are used by traders internationally. A standard bar chart shows the open, high, low, and closing price for a given time period. Candlesticks incorporate these same components but in a visually appealing format.

The body of the candlestick chart is called the real body, and represents the range between the open and closing prices. The thin vertical line above and/or below the real body is called the upper/lower shadow.

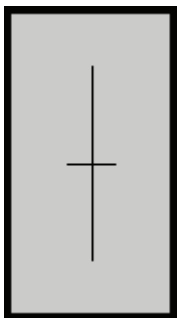
A black, red, or filled-in body represents that the close during that time period was lower than the open. A white, green, or open body represents a close that was higher than the open.

Candlestick trading signals consist of approximately 40 reversal and continuation patterns. However, most trade situations represent just a handful (10-12) of candlestick formations that the trader would benefit from committing to memory. Keep in mind that the use of these chart examples should be used as a guideline and in conjunction with the other factors of the trade such as volume, previous day's body, recent trend, etc.

DOJI

Doji are important candlesticks that provide information on their own and also feature in a number of important patterns. Doji form when a market open and close are basically equal. Alone, Doji are neutral patterns. Any bullish or bearish bias is based on previous price action and future confirmation.

A Doji can signal weakening buying pressure only when it appears after a long green candlestick or an upward trend pattern. Conversely, a Doji can signal weakening selling pressure when it appears after a long red bar or a downward trend pattern. Either way, Doji show that buying and selling pressure is evenly matched, and might indicate a possible trend reversal. This pattern alone is not enough to give the trader a sound reversal signal. One must always look for more confirmation when these Doji appear.



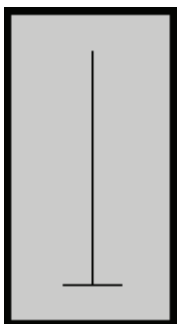
Doji: This line implies indecision. The market opened and closed at the same price.



GRAVESTONE DOJI

Gravestone Doji form when the open, low, and close are equal. The high of the day creates a long upper shadow, with an appearance similar to an upside down “T.”

Gravestone Doji show that buying pressure pushed the market higher, only to have the selling pressure push prices back to the open.

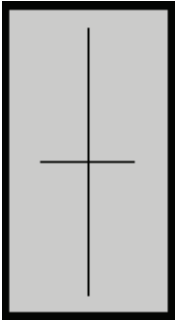


Gravestone Doji: This line indicates a turning point. It occurs when the open, close, and low are the same, and the high is higher than the open, low, and closing prices.



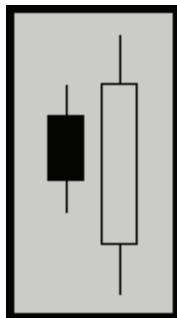
LONG-LEGGED DOJI

Long-Legged Doji show that the prices traded above and below the open, and returned to where the market opened for its settlement. This movement of higher prices and lower prices form long shadows on both sides of the open and close. This Doji could be more important after an uptrend or long green candlestick.



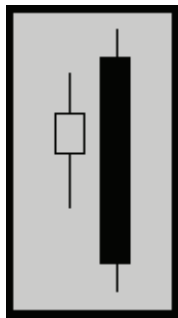
Long-Legged Doji: This line often indicates a turning point. It occurs when the open and close are the same, and the range between the high and low is relatively large.





Bullish Engulfing Lines:

This pattern is very bullish if it occurs after a fairly large downtrend acting as a reversal pattern.



Bearish Engulfing Lines:

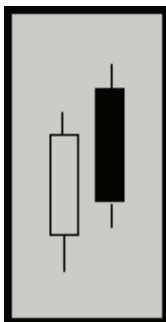
This pattern is very bearish if it occurs after a fairly large uptrend acting as a reversal pattern.

BULLISH & BEARISH ENGULFING LINES



In the Dollar Index chart above, an engulfing pattern occurs when there is an outside range day that results in a counter-trend close. What this pattern can indicate is that the market has lost the strength to continue in the direction in which it was previously headed.

DARK CLOUD COVER

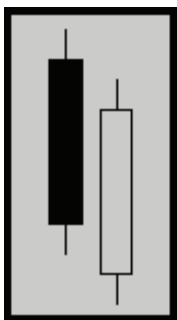


Dark Cloud Cover: This is a bearish pattern. The pattern is more important if the second line's body is below the center of the previous line's body.

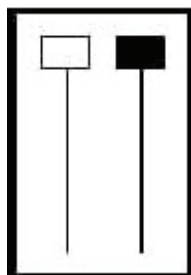


Dark Cloud Cover is a formation that suggests a market trying to reverse its trend. It is considered a bearish chart pattern. Dark Cloud Cover occurs in an up trend, when a long green candlestick is followed by a long red candlestick that opens above the prior candlestick's high. The long red candlestick must close well into the prior candlestick's range for it to be valid. This formation suggests a change in trader mindset.

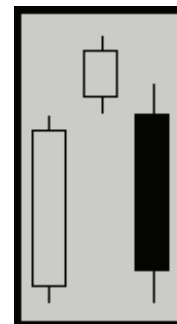
Other Candlestick Lines



Piercing Line: A bullish pattern and opposite of dark cloud cover. The open is lower than the previous low but it closes more than halfway above the first line's body.

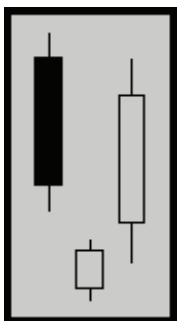


Hanging Man: These lines are bearish only if they occur after a fairly large uptrend. If this pattern occurs after a downtrend, it is called a Hammer.



Evening Star: This bearish pattern would indicate a potential top. The star indicates a possible reversal, and the bearish line confirms this.

MORNING STAR

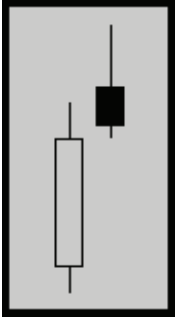


Morning Star: This bullish pattern would indicate a potential bottom. The star indicates a possible reversal and the bullish line confirms this.



The Morning Star formation occurs when you have a market with a long red body that is followed by a second green body, which gaps lower to form a star. After the star is formed you will have a third body that is a green candlestick, which closes well into the first session's red body. This hints at a change in trend without revealing the duration of this trend change.

SHOOTING STAR



Shooting Star: This pattern would suggest a minor reversal when it appears after a rally. In order for this to occur, the star's body must appear near the low price and the line should have a long upper shadow.



Shooting Stars are a bearish development. A shooting star is formed when you have a candlestick with a long upper shadow with little, if any, body that closes near the lows of the day. The body will close at the lows and this hints at a change in market sentiment. The market tries to work higher but runs out of strength and then moves into the red.

Part 4: Using and Applying Studies

There are numerous studies available that can add value to your market analysis. One of the biggest benefits to understanding technical analysis is the ability to use a combination of studies that you believe gives you an edge in the market. It is highly unlikely that you will use all of the indicators and studies available through the RJO Vantage™ online platform or the RJO Futures website. We strive to provide the most common tools available to help you become successful.

MOVING AVERAGES

A moving average is exactly as it sounds; it is the average of the sum of prices (usually the closing price) for a given number of days, divided by that number of days. It is called “moving” because the average changes each day, adding in the current day’s price and dropping the oldest day’s price. On the RJO Vantage™ charts the number of days used in calculating the moving average is 10, but that number can be altered. Ask your RJO Futures representative to walk you through editing any of the study indicators.

BOLLINGER BANDS

The Bollinger Bands are moving average lines two standard deviations above and two standard deviations below the moving average. They are plotted on the graph on either side of the moving average line. The Bollinger Bands are used to determine overbought and/or oversold areas. When the moving average approaches the upper band it is an indication of an overbought market and prices should soon drop. When the moving average line closes in on the lower band it indicates an oversold market and prices should soon rally. The width between the two bands is an indication of volatility. The farther apart the bands are from each other, the greater the volatility.

RELATIVE STRENGTH INDEX (RSI)

Relative strength, in this context, is calculated considering a given number of days (X). The average of X days that the market closed higher than it opened is divided by the average of X days that closed down. The Relative Strength Index is then calculated by dividing 100 by the quantity one plus the relative strength and subtracting 100.

$$RSI = 100 - \{100 / (1 + RS)\}$$

To interpret this index, the general rule of thumb is: If $RSI > 70$, the market is considered overbought and prices should correct and fall and if $RSI < 30$, the market is seen as oversold and prices should rise.

MOVING AVERAGE CONVERGENCE/DIVERGENCE (MACD)

The convergence/divergence analysis produces two moving average curves that move about a zero line. One line is typically a solid line and the other is dotted. The buy/sell signals occur when the two lines cross or they both cross the zero line. When the solid line crosses down through the dotted line, this indicates a sell signal. When the solid line crosses up through the dotted line it indicates a buy signal. Divergences also indicate signals. If after the solid line crosses down through the dotted line and then both lines cross the zero line, it is considered a major sell signal. The reverse movement is therefore a buy signal.

MOMENTUM

The momentum line is calculated by subtracting the current day's closing price from the closing price X number of days ago. It is used to determine overbought and oversold markets and also indicates the pace of the rise or fall of prices. When the momentum line crosses the zero line from below, it is a buy signal and when it crosses the zero line from above it is a sell signal.

WILLIAMS %R

The Williams %R is another way to detect overbought/sold markets. It is calculated by subtracting the highest price over X number of days from the price at the current day's close and then dividing that quantity by the total range (high – low) for X days. If the Williams %R is less than 20, then the market is oversold and prices should soon rise. If the %R is greater than 80, the market is overbought and prices should soon fall.

VOLUME

Volume is the total amount traded, or the number of contracts traded, of a given commodity in a single day. This number helps to determine the strength of a price trend and study indicator; the more volume in a market during a trend, the stronger or more significant the trend.

OPEN INTEREST

Open interest is the total number of open, or outstanding, contracts held by traders in a given market at the end of each day. This is calculated by counting the total long positions or short positions in the market, but not both. When considered by itself, open interest shows the liquidity of the market. If combined with volume, a rise of the two can validate a trend while a drop can indicate the end of the current trend.

COMMITMENT OF TRADERS (COT)

The Commitment of Traders report comes out weekly and breaks down the open interest into three categories: large hedgers, large speculators, and small traders. By tracking these three groups separately an analyst can see where the “smart money” is trading, because most analysts consider large traders to be the market makers.

AVERAGE DECLINE INDEX (ADI)

This index measures the strength of a market trend, not the direction as the name might lead you to believe. If the ADI is rising, the stronger the market trend has been and vice versa.

STOCHASTIC FAST & SLOW

Yet another indicator of overbought/oversold markets, the Stochastic Indicator is a way of measuring where the current day's closing price falls in with the highest highs and the lowest lows over a given period of time. There are two measurements of stochastic, %K and %D. %K is calculated by subtracting the current day's close from the lowest low over X number of days and then dividing that quantity by the range (high – low) over X days. %D is the moving average of %K. %K is often displayed as a solid line and %D dotted. If the current day closing levels are close to the lows for the given range, it is an indication of an oversold market while closing levels near the highs for the range indicate an overbought market.

Quiz Yourself: Are You Ready to Advance to the Next Step or Do You Need to Review?

1. Which of the following is true?

- a. Up trendlines should be viewed as support until broken
- b. Down trendlines should be viewed as resistance (or a ceiling) until broken
- c. Both a and b
- d. Neither a nor b

2. In the pit traded markets, when is volume posted?

- a. In real time
- b. With a one hour lag
- c. With a one day lag
- d. With a one week lag

3. Which of these factors come into play when identifying a trend change?

- a. The market closes above the resistance or below the support trendline
- b. Volume tends to increase as a trendline is broken
- c. The simple moving average changes direction or the exponential moving average lines cross over
- d. None of the above
- e. All of the above

4. Which of these is created as a result of a sharp high-volume market move and the need for the market to take a breather?

- a. A flag
- b. A pennant
- c. Both a and b
- d. Neither a nor b

5. Ascending triangle patterns are considered

- a. Bullish
- b. Bearish
- c. Neutral
- d. None of the above

6. In which pattern does volume remain higher?

- a. Rectangle
- b. Triangle
- c. Volume is equal in both

7. Which of the following is true?

- a. A falling wedge is considered bearish
- b. A rising wedge is considered bearish
- c. Neither a nor b
- d. Both a and b

8. Prior to a continuation gap, a market will congest for a period of time ranging from days to weeks.

- a. True
- b. False

9. Which of these signals that the market is preparing for a major trend change?

- a. Tops
- b. Bottoms
- c. Neither a nor b
- d. Both a and b

10. Where does candlestick charting originate from?

- a. India
- b. China
- c. Japan
- d. The United States

11. How many reversal and continuation patterns do candlestick trading signals consist of?

- a. 5
- b. 20
- c. 40
- d. 100

12. When do Doji candlesticks form?

- a. When a market open and close are basically equal
- b. When a market opens high and closes low
- c. When a market opens low and closes high
- d. None of the above

13. A Dark Cloud Cover formation is considered

- a. Bullish
- b. Bearish
- c. Neither a nor b

14. A Piercing Line formation is considered

- a. Bullish
- b. Bearish
- c. Neither a nor b

Quiz Yourself: (Continued)

15. Which formation is Chart A an example of?

- a. A double top
- b. A double bottom
- c. A morning star
- d. A shooting star
- e. None of the above

16. Which formation is Chart B an example of?

- a. A symmetrical triangle
- b. An island reversal
- c. A wedge
- d. A morning star
- e. None of the above

17. A shooting star formation is considered

- a. Bullish
- b. Bearish
- c. Neither a nor b

18. A moving average is the average of the sum of prices for a given number of days, divided by that number of days.

- a. True
- b. False

19. Bollinger Bands are moving average lines three standard deviations above and three standard deviations below the moving average.

- a. True
- b. False

20. Volume helps to determine the strength of a price trend and study indicator.

- a. True
- b. False

CHART A



CHART B





Quiz Yourself: (Continued)

SCORING (OUT OF 20 POSSIBLE POINTS)

17-20 = You're Technically Savvy.

You may be ready to trade. Contact an RJO Futures representative at 800-441-1616 now, and learn how you can transform your knowledge of Technical Analysis into possible trading opportunities.

12-16 = You May Want to Revisit the Material.

You've learned a fair amount about technical analysis. But we recommend you revisit the material to fully grasp the concepts. Once you have it down, you may be ready to apply what you've learned to your trading.

1-11 = Definitely Revisit the Material, and Take the Quiz Again.

You simply need to reread the material and/or contact an RJO Futures Trading consultant at 800-441-1616 for assistance. We'll be happy to walk you through any parts of this guide to help you to better understand the content. And we offer many other resources to help you along the way.

1 (c), 2 (c), 3 (e), 4 (c), 5 (a), 6 (a), 7 (b), 8 (a), 9 (c), 10 (c), 11 (c), 12 (a), 13 (b), 14 (a), 15 (b), 16 (c), 17 (b), 18 (a), 19 (a), 20 (a)

Each correct answer equals 1 point.

My score: _____

More Information About Technical Analysis

This intro to technical analysis is meant to be just that, an intro. In order to take advantage of the full potential value of technical analysis, we encourage you to learn more about it.

As a next step, we invite you to contact one of our Trading Advisors here at RJO Futures. He or she will be able to walk you through some of the principles detailed in this intro - as well as take you to the next level in your understanding of technical analysis and its uses.

Contact us at:

Phone: (800) 441-1616 or (312) 373-5478

Email: info@rjofutures.com

Web: www.rjofutures.com

Twitter: www.twitter.com/rjofutures

ADDITIONAL RESOURCES

RJO Futures Market News, E-newsletter

This bimonthly newsletter features market analysis, reports, and commentary from our trading strategists.

<http://www.rjofutures.com/eview/>

RJO Futures Intro to Fundamental Analysis

Now that you've got a primer on Technical Analysis, why not give our *Intro to Fundamental Analysis* guide a try?

Contact an RJO Futures Trading Strategist at (800) 441-1616 or (312) 373-5478 to get your free copy today.

RJO Futures Risk Management

A successful trading plan includes a sound risk management plan.

Contact an RJO Futures Trading Strategist at (800) 441-1616 or (312) 373-5478 to get your free guide today.